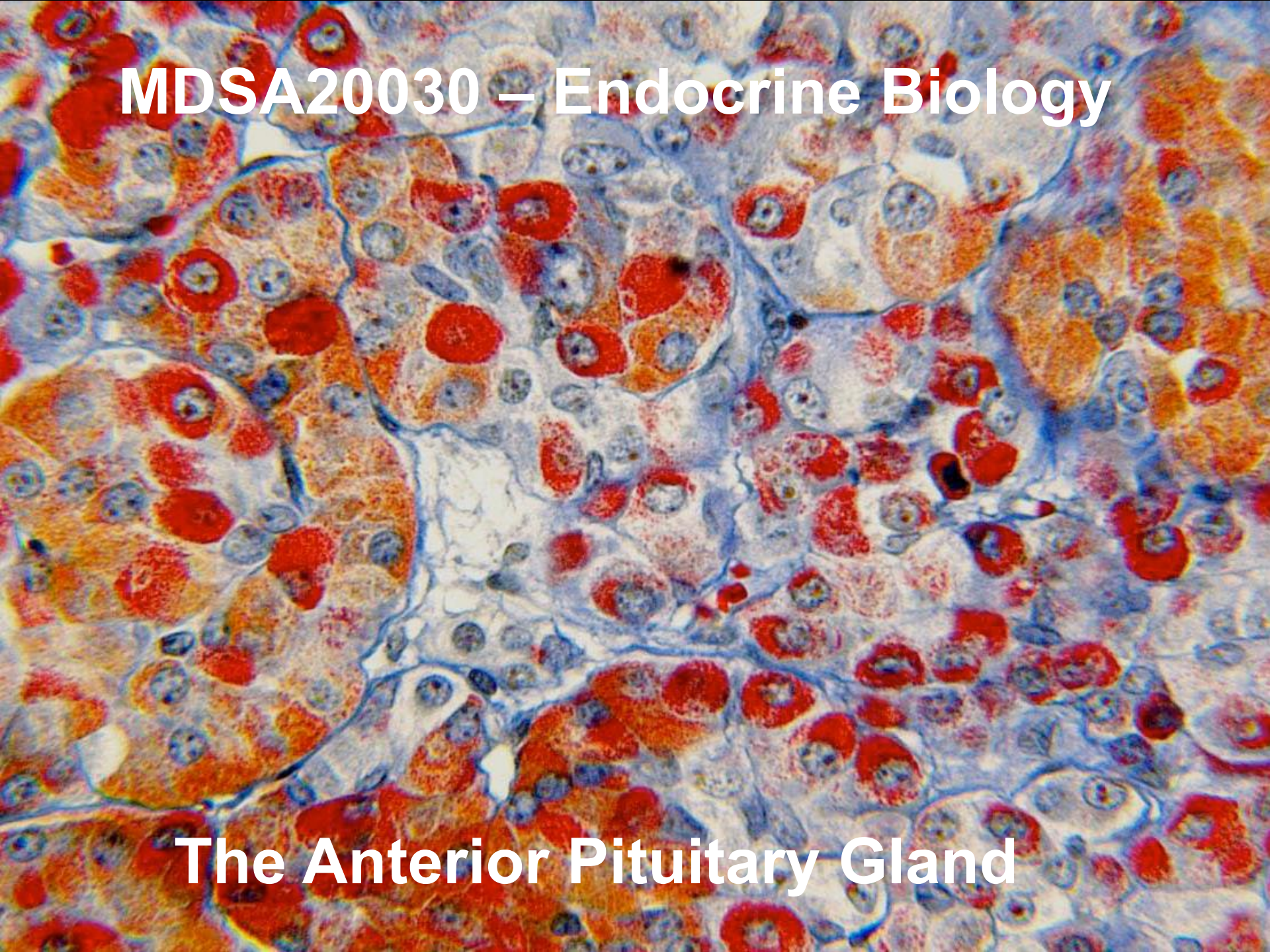


# MDSA20030 – Endocrine Biology

A high-magnification light micrograph of a histological section of the anterior pituitary gland. The image displays a variety of cell types, including somatotrophic, lactotrophic, and gonadotrophic cells. These cells are characterized by their large, pale-staining nuclei and prominent, eosinophilic (pink) secretory granules. The cells are arranged in a somewhat disorganized pattern, with some cells showing more granules than others. The overall color palette is dominated by the pink of the granules and the blue of the nuclei, with some areas of lighter, more eosinophilic cytoplasm.

## The Anterior Pituitary Gland

# Learning Objectives

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- After this session, you should have a greater understanding of
  - the embryological origins of the anterior pituitary and
  - the developmental pathways giving rise to the endocrine cells of the anterior pituitary
  - endocrine functions of the anterior pituitary
  - pituitary dysfunction (hypopituitarism and pituitary adenomas)
  - prolactin physiology

# Biochemistry Practical

15:00 – 17:00 in Science Centre East (E2.53 and 2.54)

Only attend the session you are assigned to – consult the listings on Brightspace (Laboratory Practical –GTT folder)

Bring lab coat, graph paper and writing materials

Print out and partially fill in a lab report before the practical (title, background, aims, methods) – leave some blank space to enter your results you will complete it on the day

Please read the supplementary material in the Brightspace folder

For those wishing to do their own personal GTT, please print out and sign the consent form and bring it to the practical

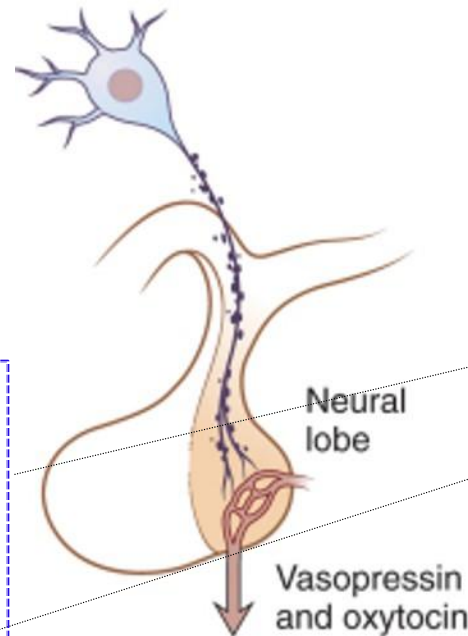
# Hypothalamic Neuroendocrine Cells

- Neuroendocrine cells
  - neurons that release peptides into the circulation 'neuroendocrine hormones'
- Magnocellular (large, long) neurons
  - large cell bodies with long axons projecting down to (posterior) pituitary
  - oxytocin and vasopressin (ADH)
  - released into systemic circulation via vasculature of posterior pituitary
- Parvocellular neurons (small, short)
  - small cell bodies with shorter axons projecting to top of the pituitary stalk (infundibulum)
  - TRH, CRH, SST, GHRH, GnRH, dopamine
  - released into hypophyseal portal circulation

Some nuclei have both magno- and parvocellular neuroendocrine cells

## Magnocellular Neuron

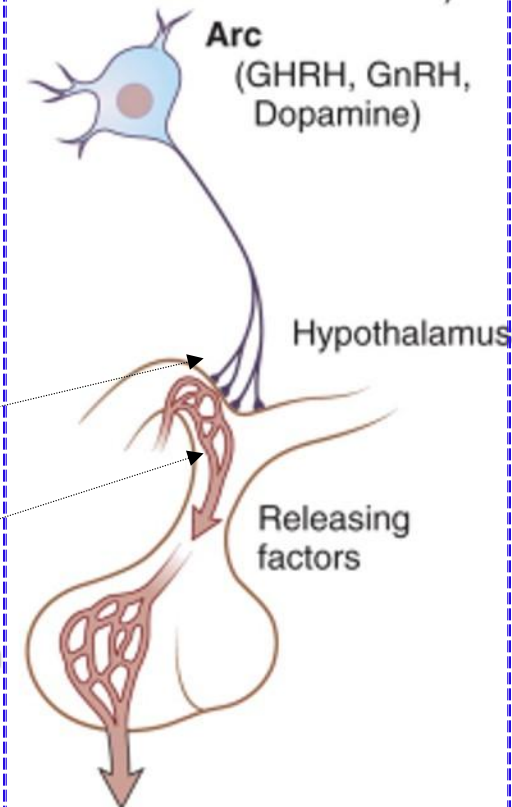
Location: SON, PVN  
(AVP, OXY)



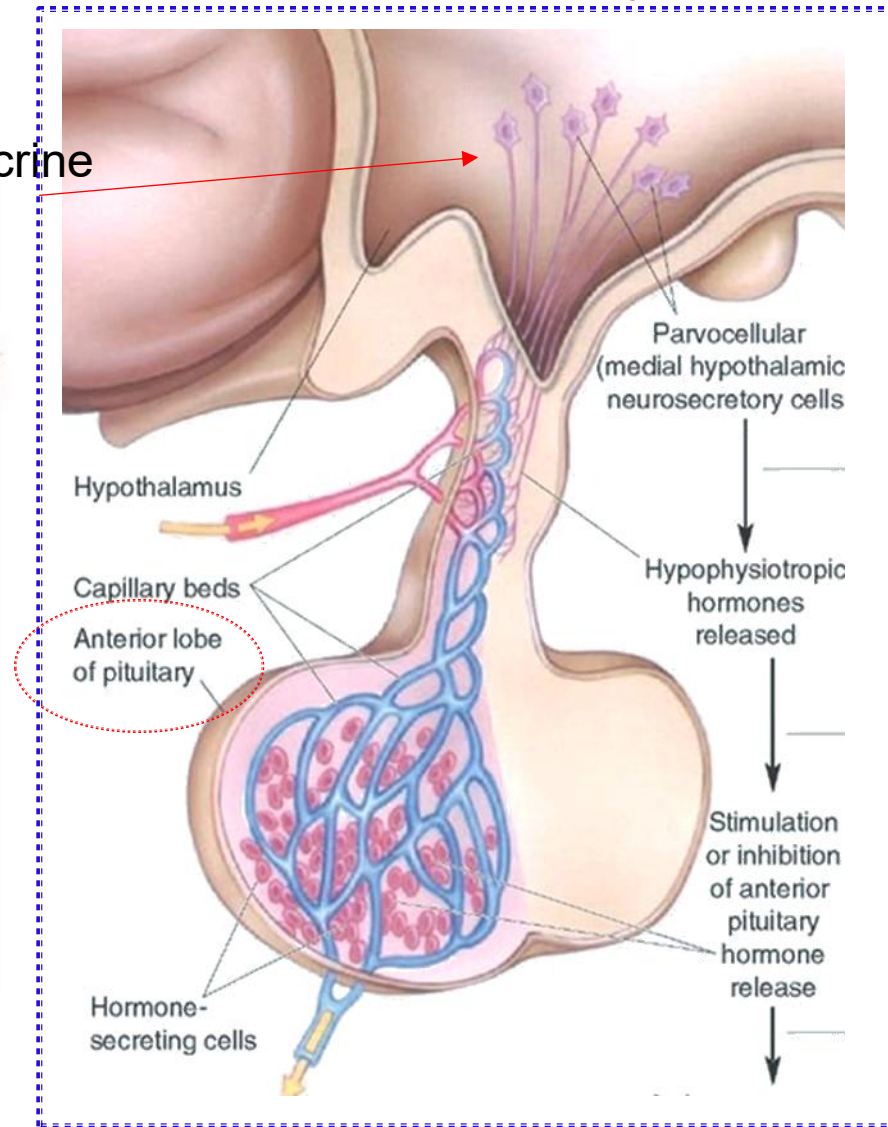
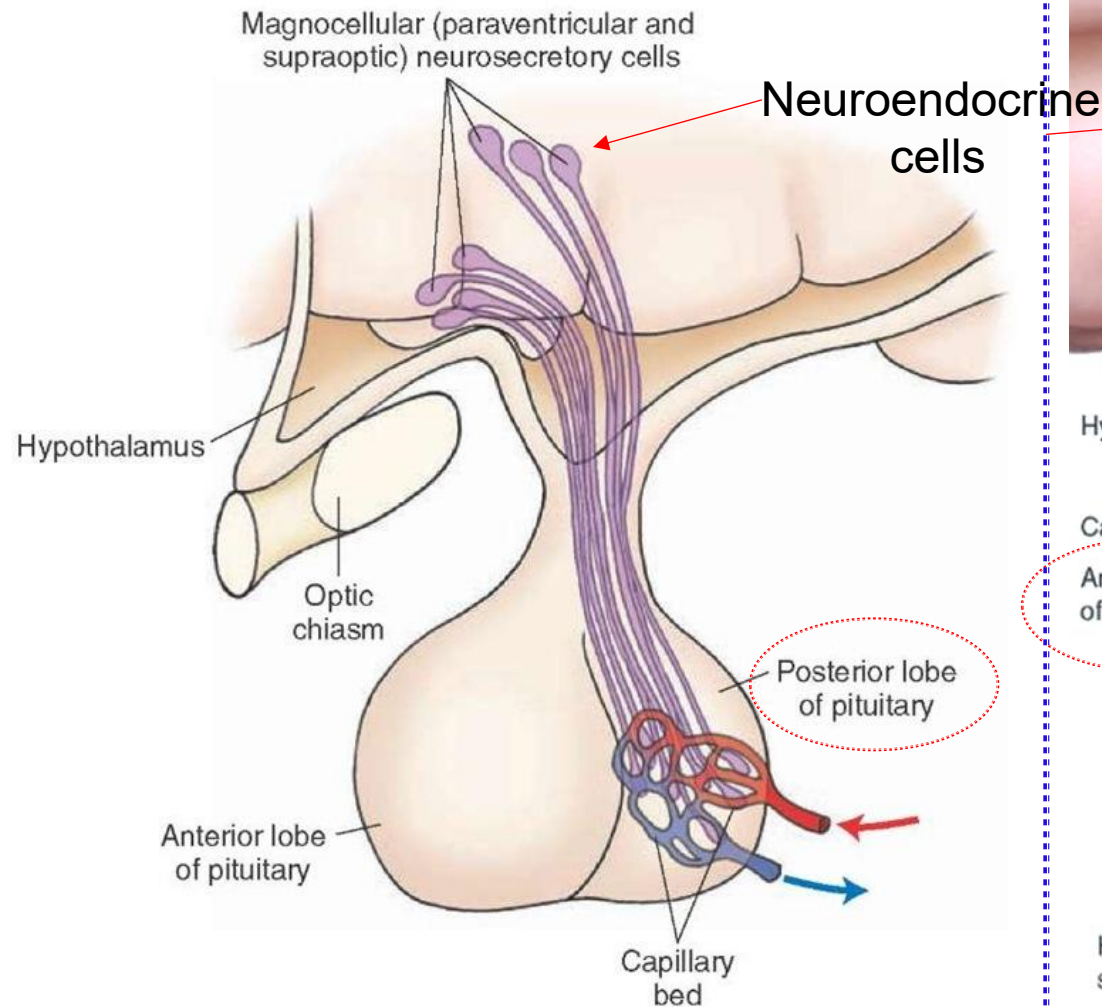
## Parvocellular Neuron

Location: PVN  
(TRH, CRH, Somatostatin)

Arc  
(GHRH, GnRH, Dopamine)



# Efferent Outputs from the Hypothalamus to the Pituitary Gland



Neuroendocrine cells in hypothalamus release hormones directly into systemic circulation (posterior pituitary) OR they release them into a local portal circulation system that perfuses the anterior pituitary and controls hormone secretion from it

# Nomenclature

## Adenohypophysis

| Anterior pituitary |  |
|--------------------|--|
| Pars tuberalis     |  |
| Pars intermedia    |  |
| Pars distalis      |  |

Hypophyseal fossa  
in sella turcica of  
sphenoid bone

Optic  
chiasm

Hypothalamus

Mammillary body

Median eminence

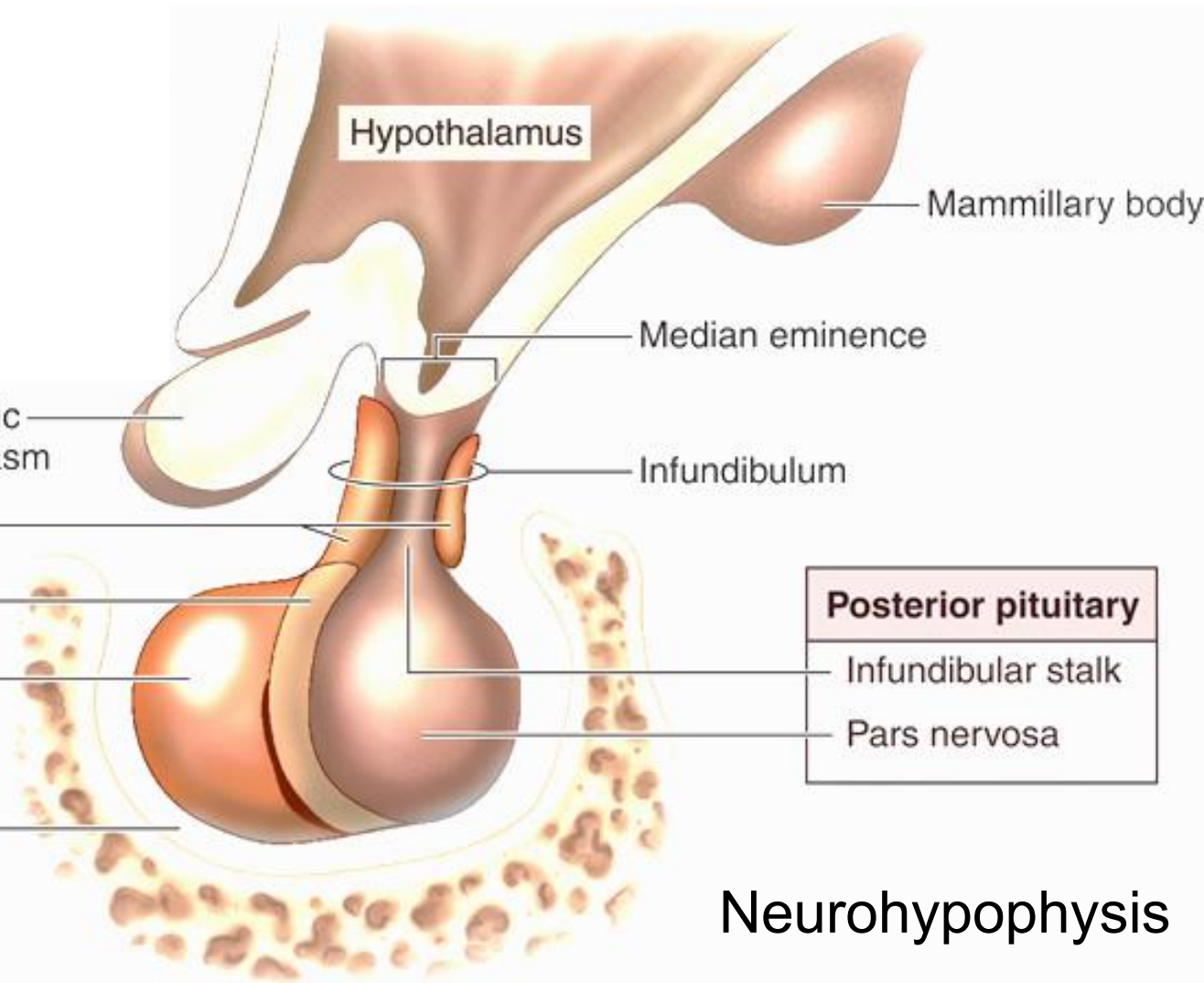
Infundibulum

Posterior pituitary

Infundibular stalk

Pars nervosa

Neurohypophysis

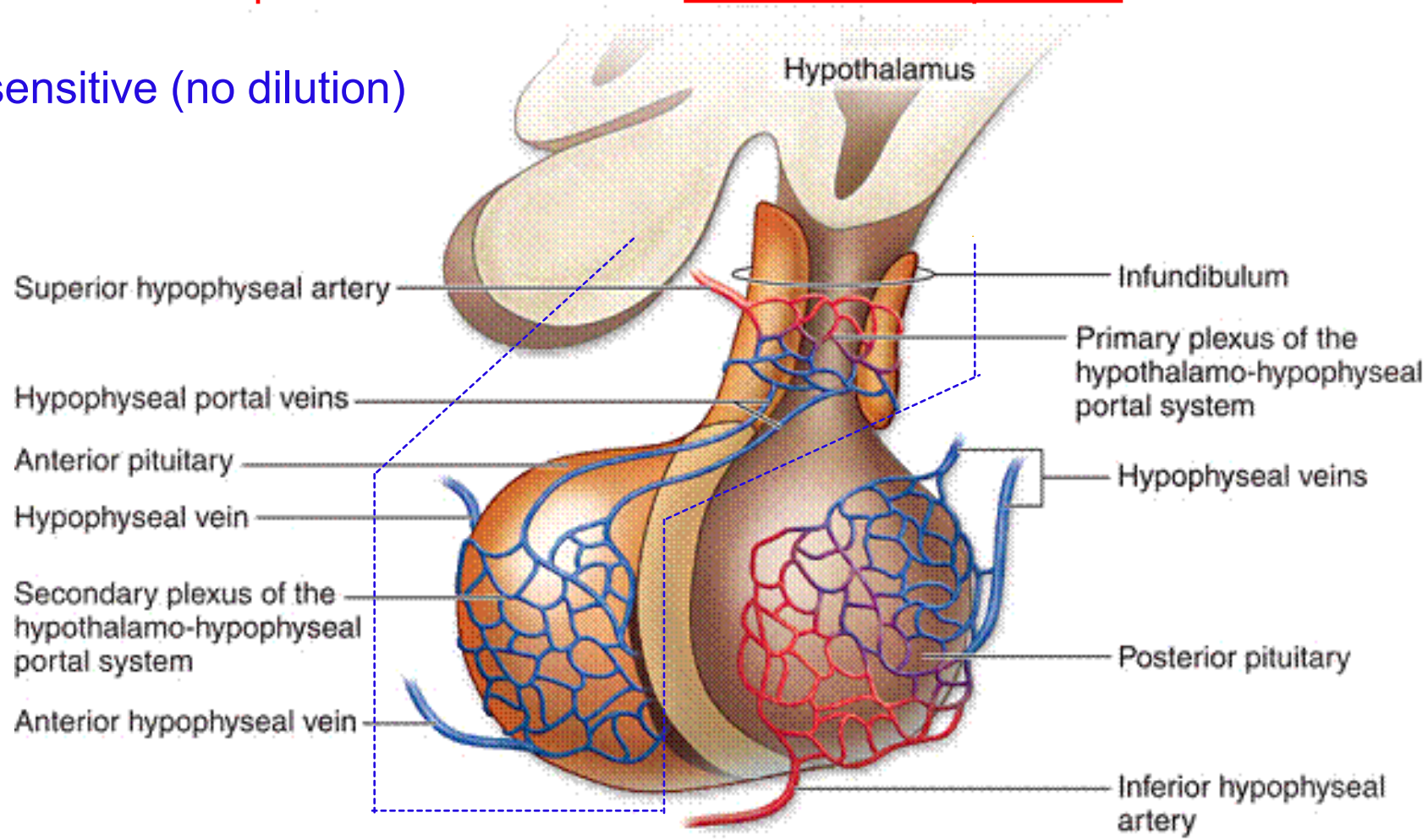


# Hypophyseal Portal Circulation System

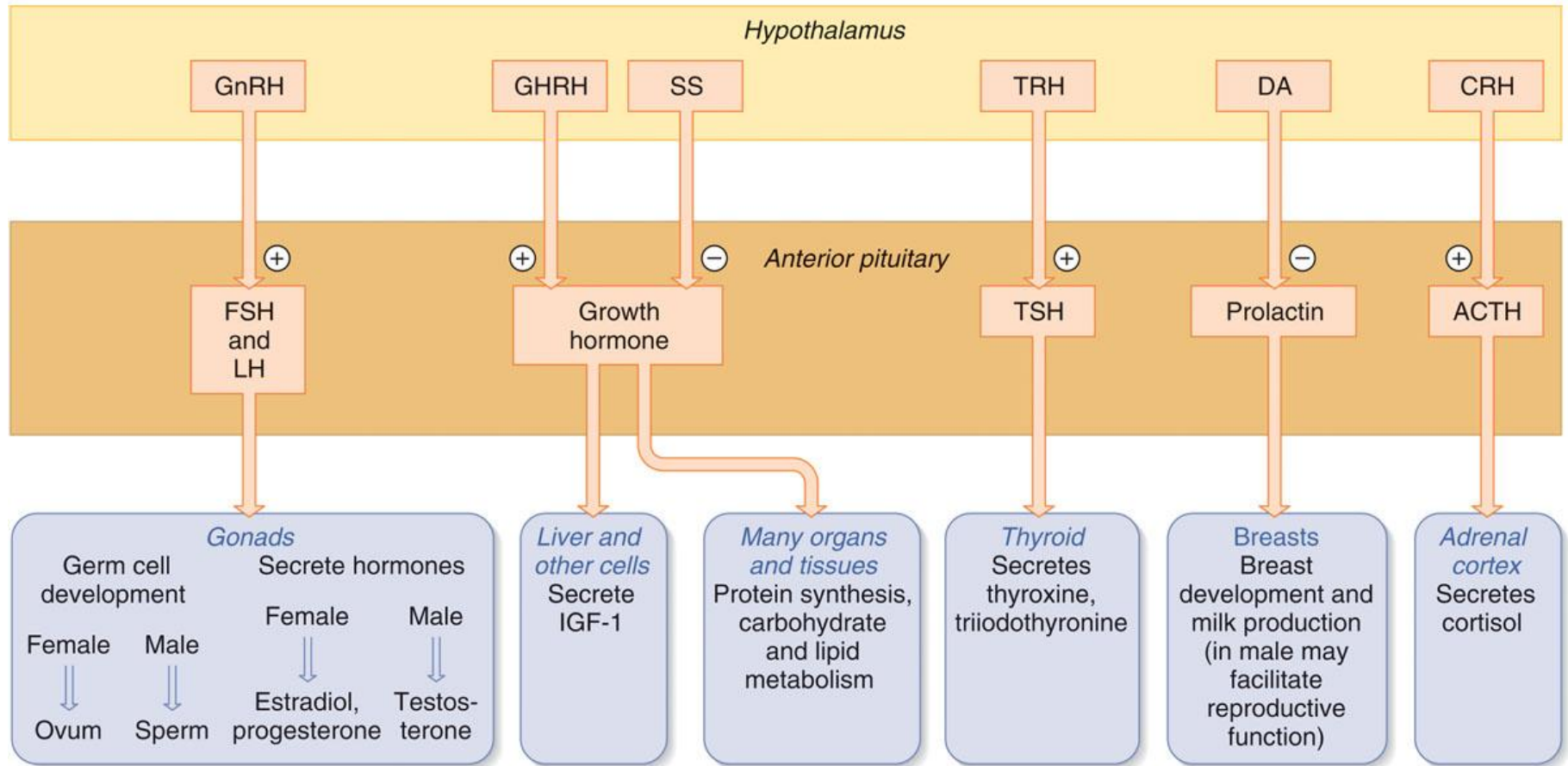
Hypothalamus/pituitary is one of the few areas of the brain where the blood brain barrier is compromised because of fenestrated capillaries

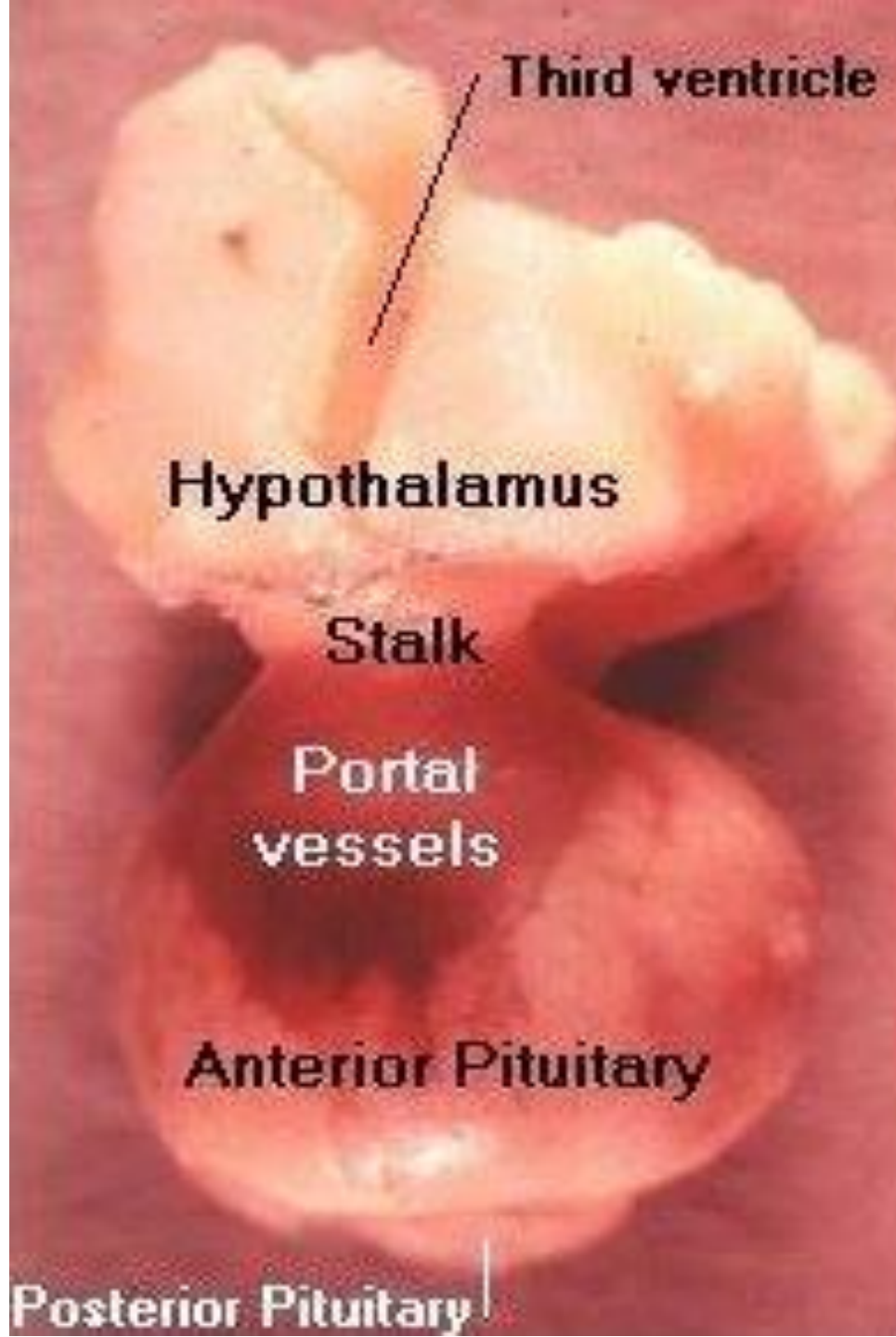
More sensitive (no dilution)

Rapid

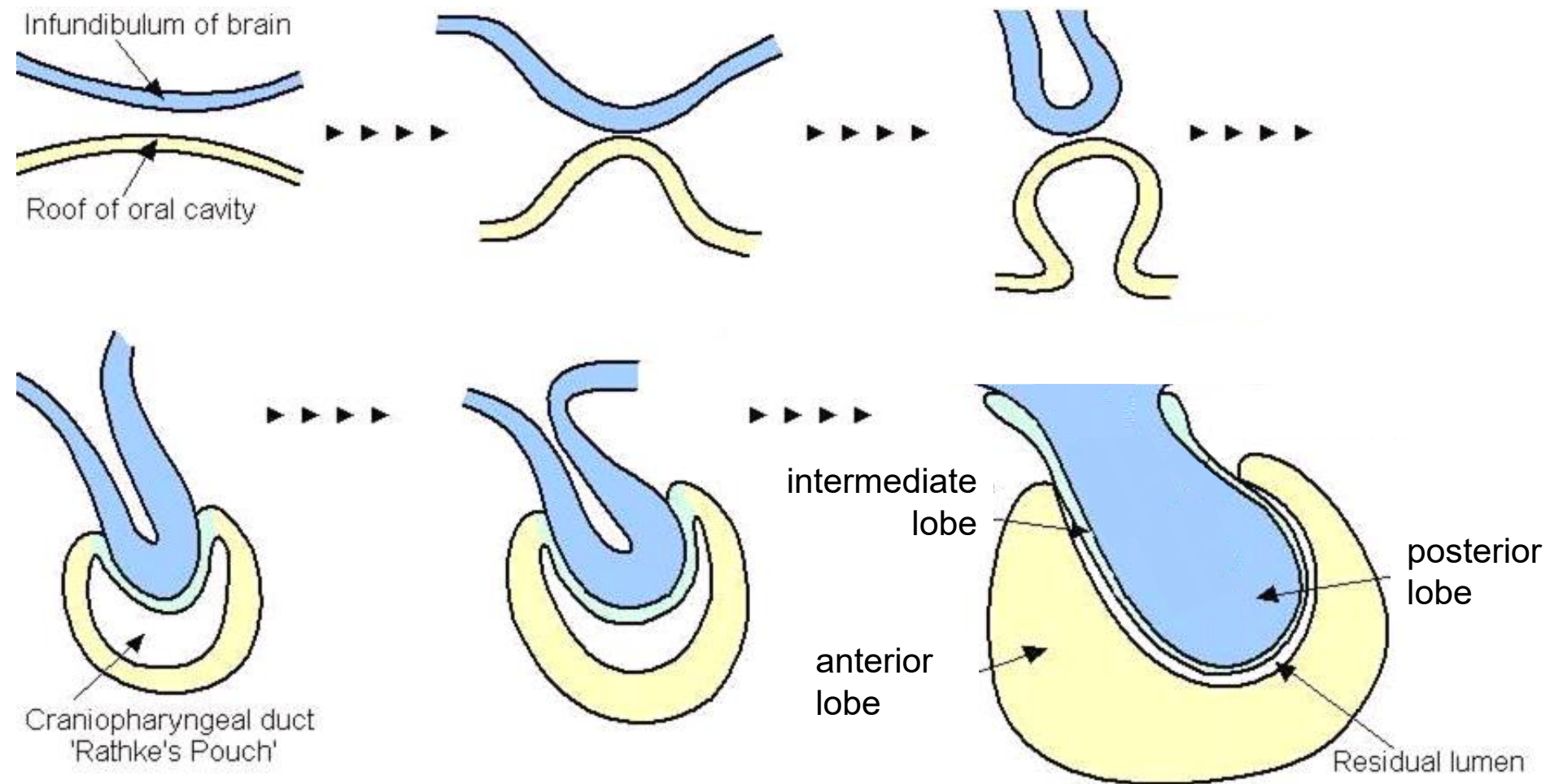


# Regulation of Anterior Pituitary Function by the Hypothalamus

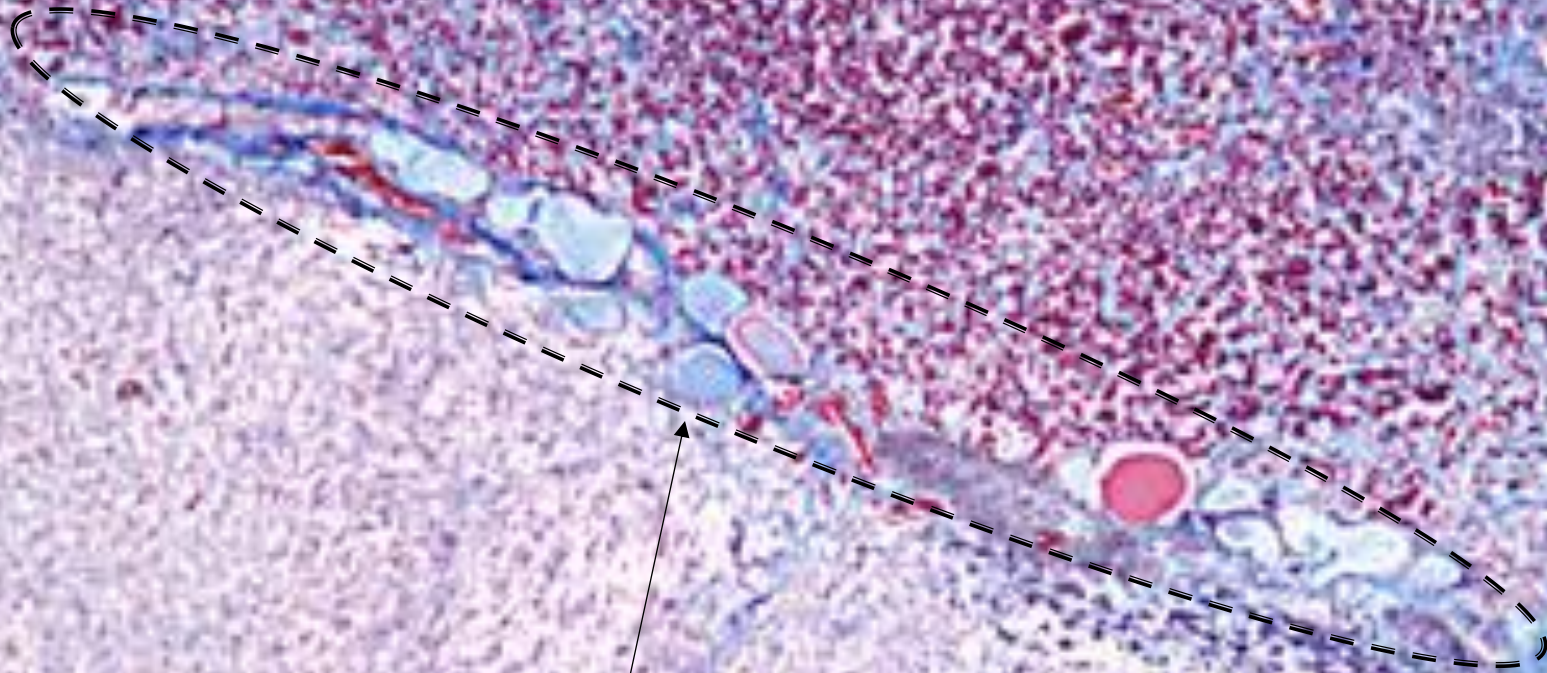




# Pituitary Development

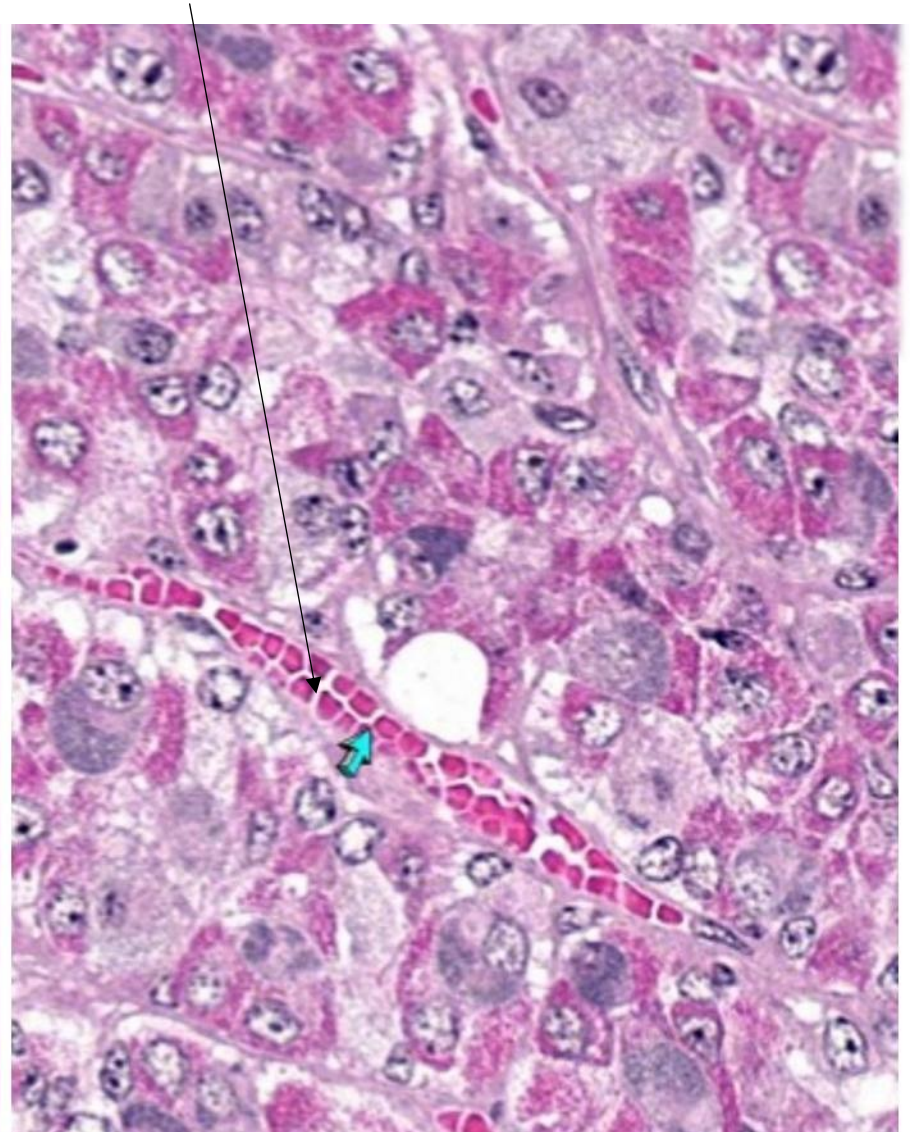
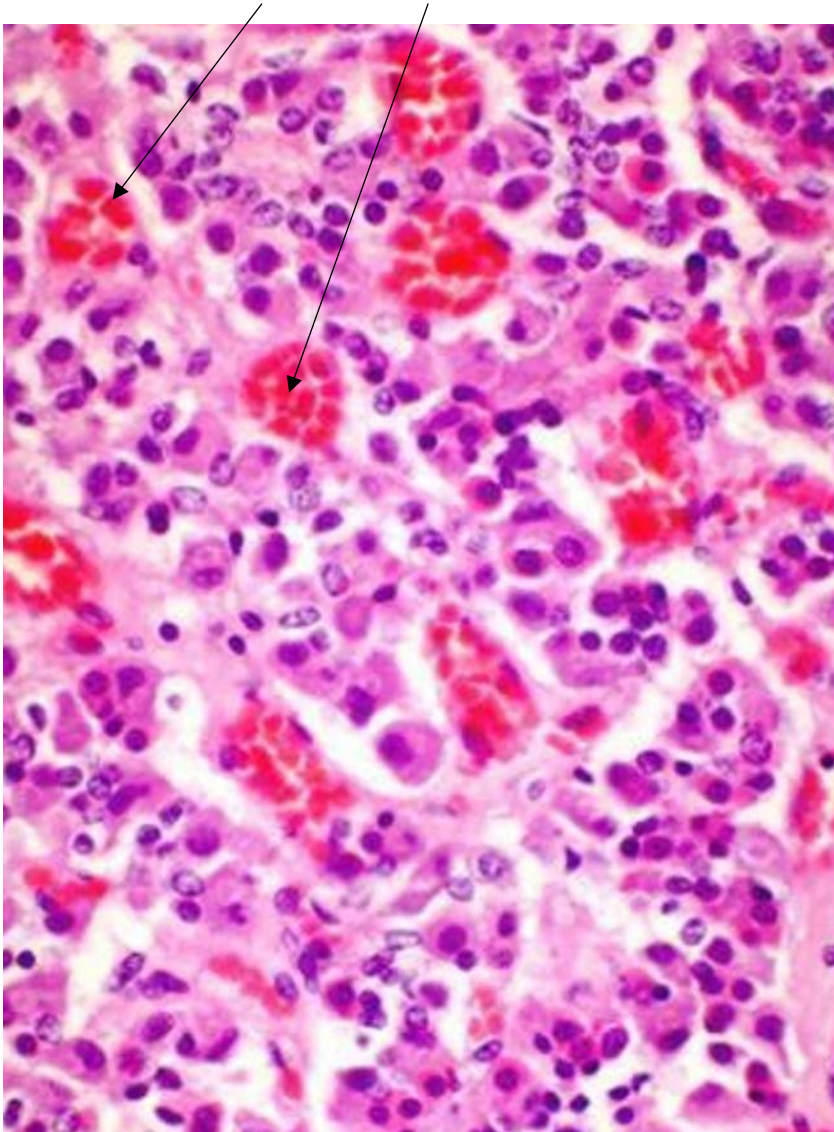


# Pars Intermedia

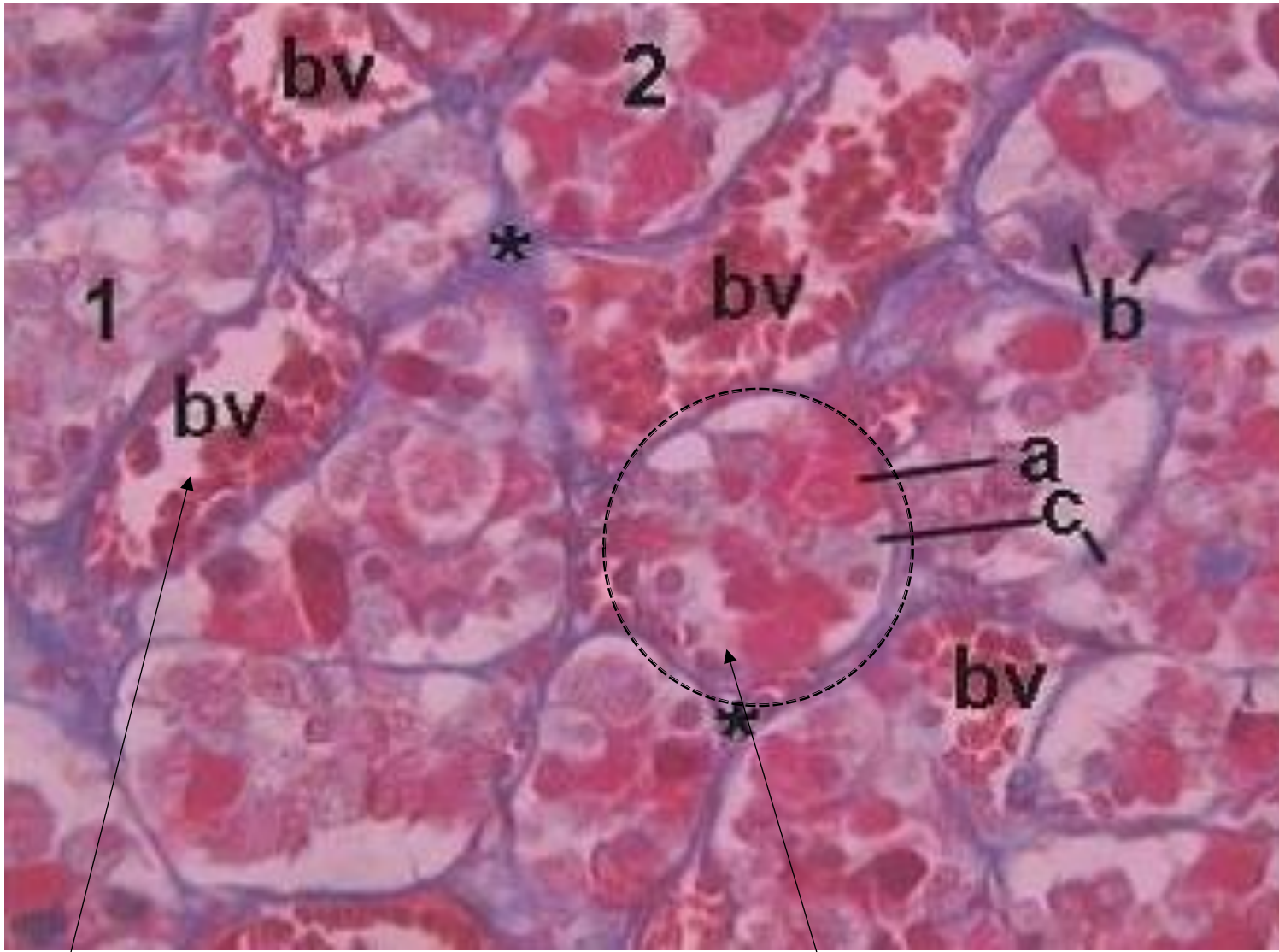


During development a transitional (intermediate) zone is formed between the neurally derived posterior lobe & the epithelially derived anterior lobe. A characteristic feature of this region are colloid filled cysts which are in fact the remnants of Rathke's pouch. It is lost in adults of some species like humans but persists in others. It makes melanocortin (MSH).

# Capillaries / Blood Sinuses in the Anterior Pituitary



# .... at Higher Magnification



A – acidophiles  
B – basophiles  
C - chromophobes

clusters of endocrine cells enclosed by connective tissue (blue stain)

adjacent capillaries/sinuses (bv)

# Anterior Pituitary Endocrine Cell Types

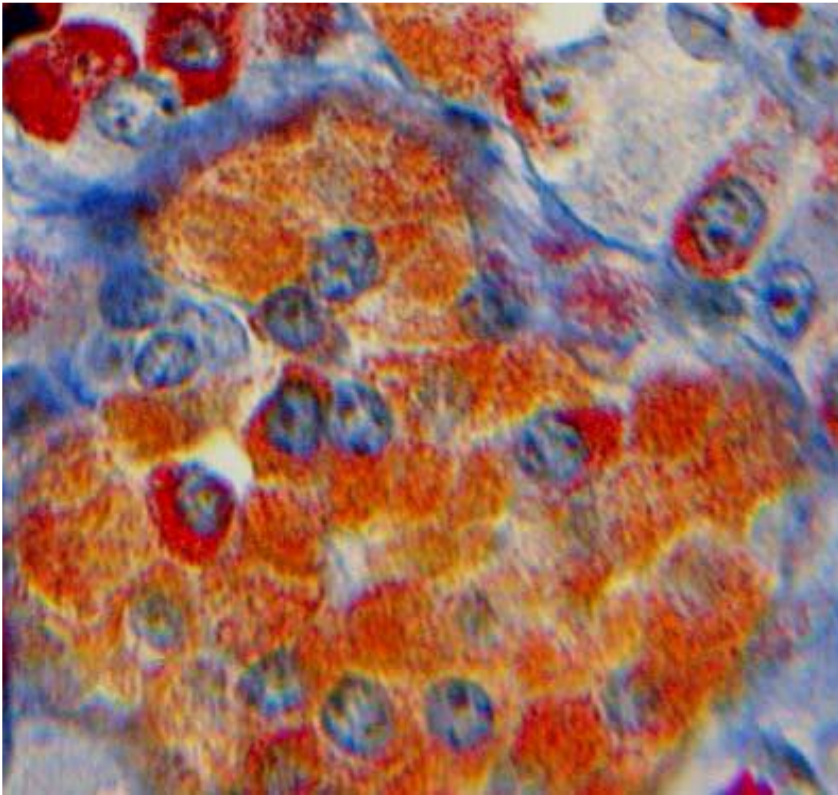
Acidophiles – somatotrophs and lactotrophs

Basophiles – gonadotrophs, corticotrophs, thyrotrophs

Chromophobes – ‘retired’ endocrine cells (stain poorly)

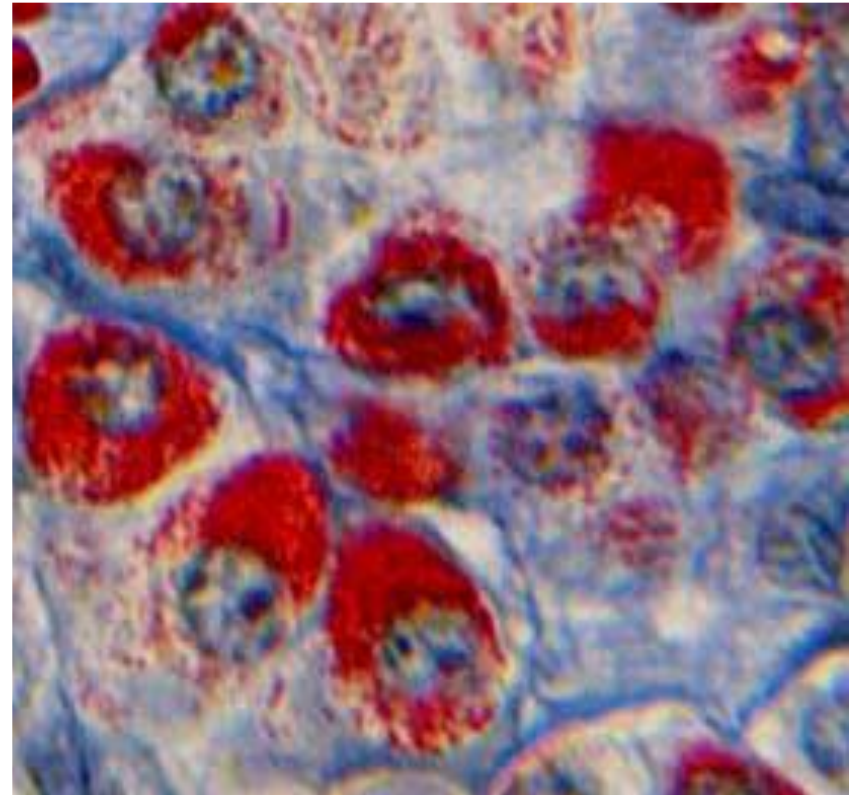
| Cell name     | Relative proportion | Hypothalamic regulators | Hormone produced | Target Gland                 |
|---------------|---------------------|-------------------------|------------------|------------------------------|
| Corticotrophs | 15-20%              | CRH                     | ACTH             | Adrenal gland                |
| Gonadotrophs  | 10-15%              | GnRH                    | FSH and LH       | Gonads                       |
| Thyrotrophs   | 5%                  | TRH                     | TSH              | Follicular cells             |
| Lactotrophs   | 15-25%              | Dopamine                | Prolactin        | Mammary glands               |
| Somatotrophs  | 35-45%              | GHRH / SST              | GH               | Liver, bone, skeletal muscle |

# Acidophiles



## Somatotrophs

Large cells in clusters  
Large secretory granules  
Growth hormone



## Lactotrophs

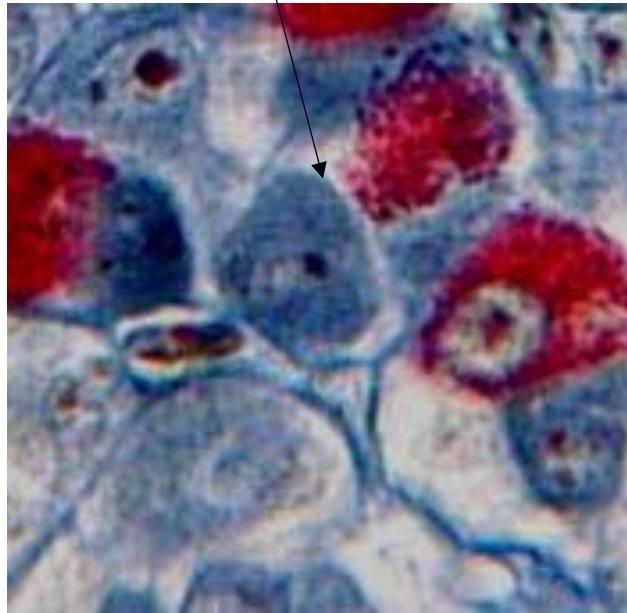
Solitary cells with eccentric nuclei  
Large secretory granules  
Prolactin

# Basophiles

## Gonadotrophs

Small round cells with strong basophilic appearance

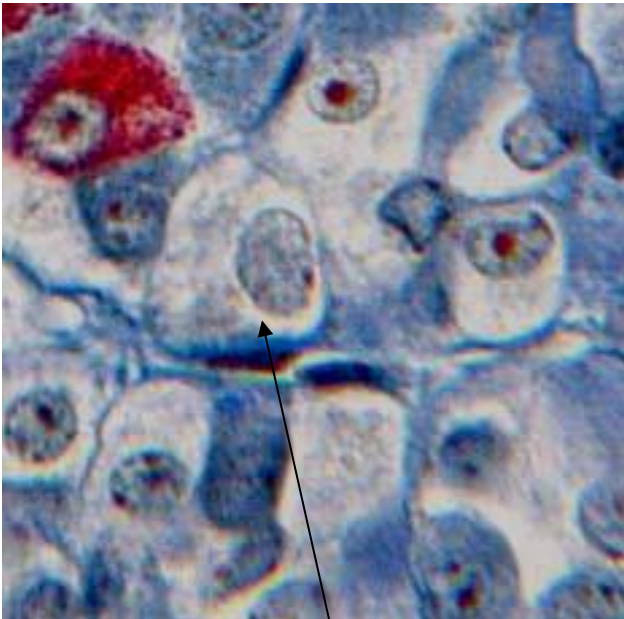
FSH and LH



## Thyrotrophs

Large, pale, polyhedral cells  
Irregular or angular in shape

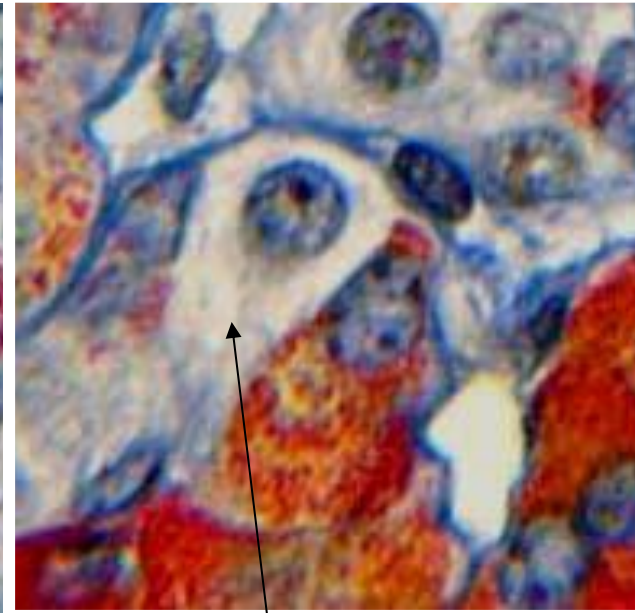
TSH



## Corticotrophs

Pale cells with weak basophilic appearance. Cells may be spherical, ovoid or stellate

ACTH



# Pituitary Adenomas

| Tumor                               | Hormones                                     | Transcription factor        |
|-------------------------------------|----------------------------------------------|-----------------------------|
| Somatotroph adenomas                |                                              |                             |
| Densely granulated                  | GH, $\alpha$ -subunit                        | Pit-1                       |
| Sparsely granulated                 | GH ( weak)                                   |                             |
| Mammotroph adenomas                 | GH, PRL, $\alpha$ -subunit                   |                             |
| Mixed GH- PRL adenomas              | GH,PRL, $\alpha$ -subunit                    |                             |
| Plurihormonal GH-producing adenomas | GH, PRL $\alpha$ -subunit, $\beta$ -TSH      |                             |
| Lactotroph adenomas                 |                                              |                             |
| Sparsely granulated                 | PRL, $\alpha$ -subunit                       | Pit-1                       |
| Densely granulated                  | PRL                                          |                             |
| Acidophil stem cell adenomas        | PRL, GH                                      |                             |
| Thyrotroph adenomas                 | $\beta$ -TSH, $\alpha$ -subunit              | Pit-1, GATA-2               |
| Adenomas producing ACTH             |                                              |                             |
| Densely granulated                  | ACTH                                         | Tpit                        |
| Sparsely granulated                 | ACTH                                         |                             |
| Crooke's cell adenoma               | ACTH                                         |                             |
| Adenomas producing FSH, LH          |                                              |                             |
| Gonadotroph adenomas                | $\beta$ -FSH, $\beta$ -LH, $\alpha$ -subunit | SF-1, ER- $\alpha$ , GATA-2 |

Can cause visual field disturbances due to compression of optic nerve

Functioning adenomas – secrete hormone in excess

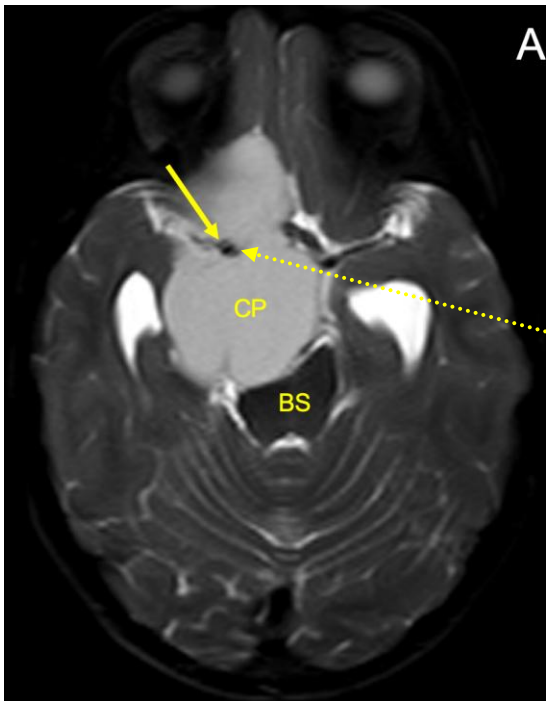
Non functioning adenomas – benign/non secretory but can cause hypopituitarism due to mass effects, compression of adjacent structures, necrosis

# Craniopharyngiomas

Craniopharyngiomas are rare, usually benign slow growing epithelial tumours, mainly of the suprasellar region. Most extend towards the hypothalamus and optic chiasm

Arise from epithelial remnants associated with development of **Rathke's pouch**. Although benign, their location can cause substantial pituitary, hypothalamic and visual dysfunction.

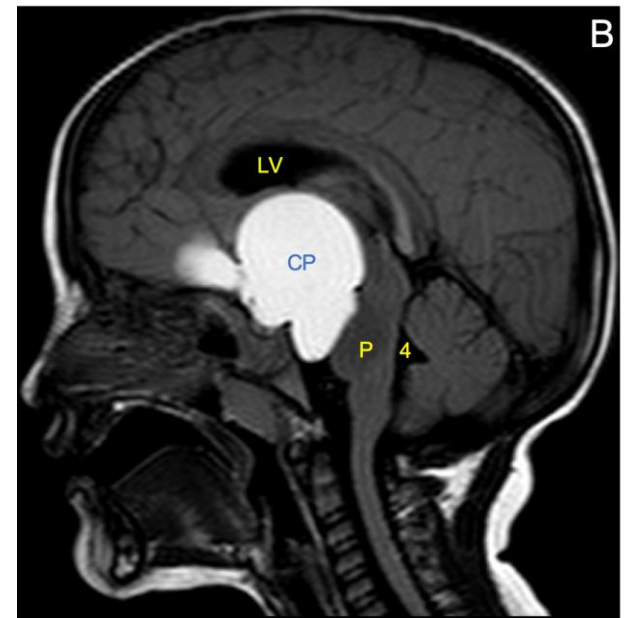
Axial T2-weighted MRI



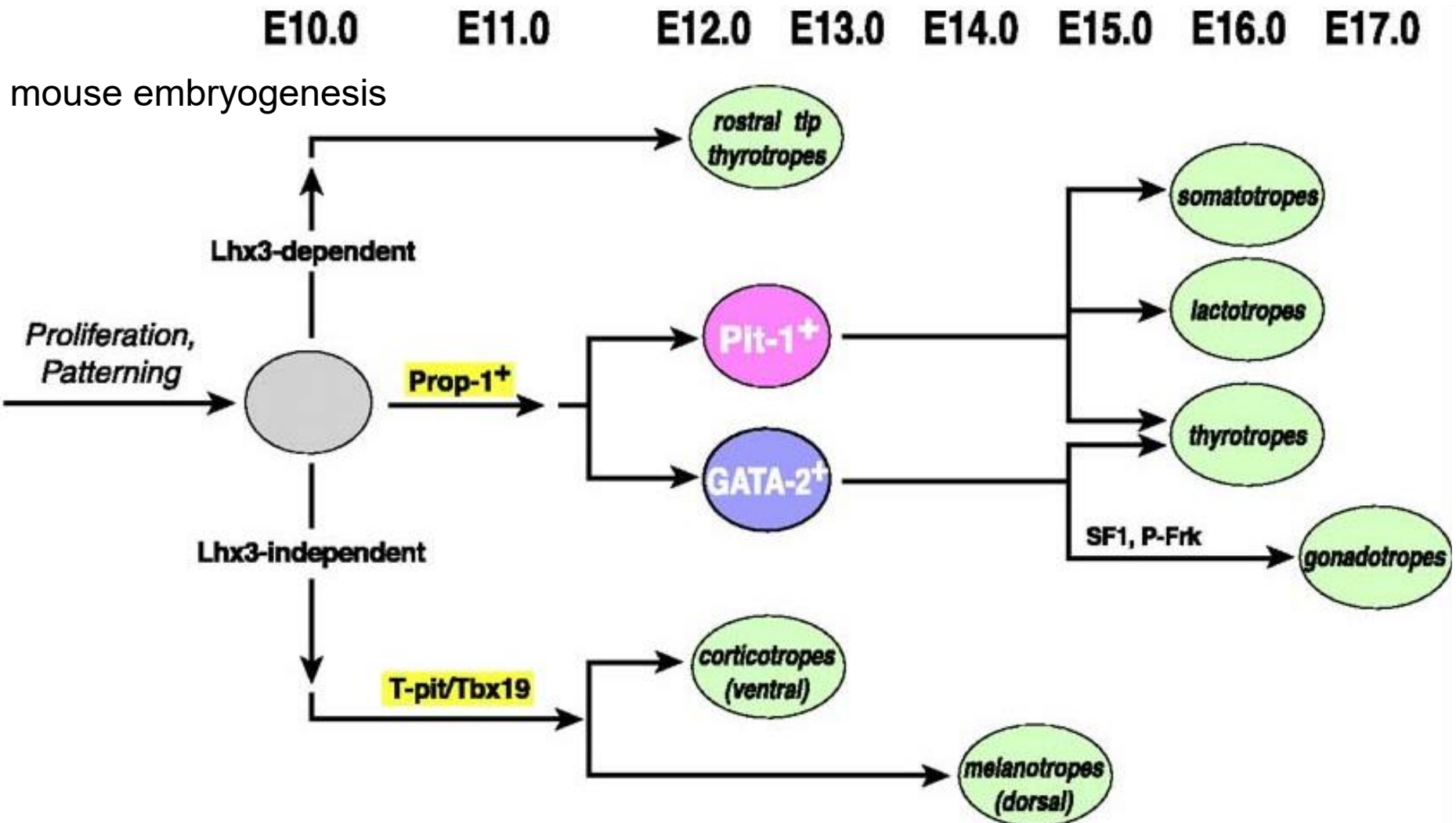
CP = Craniopharyngioma  
BS = Brainstem  
LV = lateral ventricle  
P = pons  
4 = fourth ventricle

Optic nerve

Mid-sagittal MRI

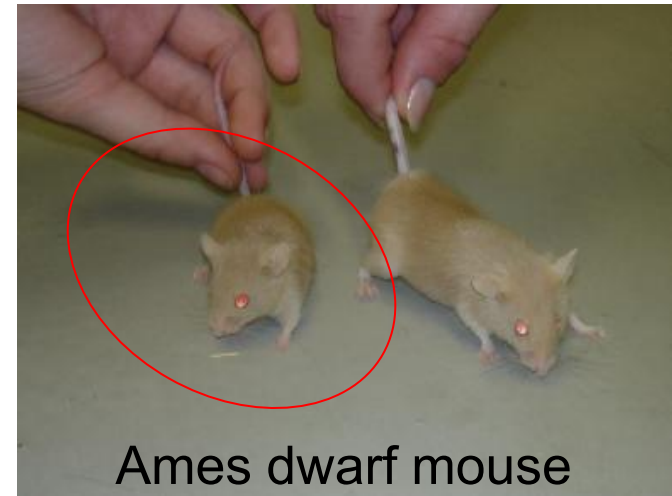


# Differentiation of Anterior Pituitary Endocrine Cells

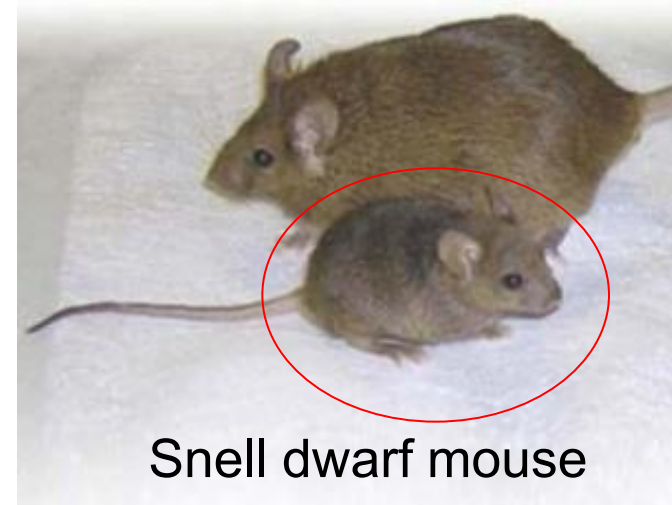


# Failure of Anterior Pituitary Development

- Ames dwarf mouse
  - recessive PROP1 mutation
  - ~ 30% of normal size
  - ↓ prolactin, GH, TSH, LH and FSH
  - sub- / infertile
  - increased longevity (50%) !!!
- Snell dwarf mouse
  - recessive PIT1 mutation
  - ↓ prolactin and GH (and TSH)
  - similar phenotype to Ames mice
  - increased longevity (50%) !!!
- Some human hypopituitary syndromes can be caused by mutations in same genes



Ames dwarf mouse



Snell dwarf mouse

# Anterior Pituitary Hormones

| Hormone                      | Other names   | Symbol | Secretory cells                        | Target              |
|------------------------------|---------------|--------|----------------------------------------|---------------------|
| Adrenocorticotrophic hormone | Corticotropin | ACTH   | Corticotrophs                          | Adrenal gland       |
| Luteinizing hormone          | Lutropin      | LH     | Gonadotrophs                           | Gonads              |
| Follicle-stimulating hormone |               | FSH    | Gonadotrophs                           | Gonads              |
| Growth hormone               | Somatotropin  | GH     | Somatotrophs                           | Liver, bone, muscle |
| Prolactin                    |               | PRL    | Lactotrophs, also known as Mammotrophs | Mammary glands      |
| Thyroid-stimulating hormone  | Thyrotropin   | TSH    | Thyrotrophs                            | Thyroid gland       |

- All are peptide hormones and all are trophic hormones except prolactin
- Hormone release is a receptor mediated process ie. the hypophysiotrophic hormones bind to cell surface receptors on AP endocrine cells, which ultimately triggers hormone release

# Classification of Anterior Pituitary Hormones

## The Melanocortins

Adrenocorticotrophic hormone (ACTH)

Lipotropin (LPH)

Melanocyte-stimulating hormone (MSH)

all derived from common precursor (POMC)

## The Somatomammotropins

Growth hormone (GH)

Prolactin (PRL)

Placental lactogen (PL)

## The Glycoprotein Hormones

Luteinizing hormone (LH)

Follicle-stimulating hormone (FSH)

Thyroid-stimulating hormone (TSH)

- dimers  $\alpha$  and  $\beta$  subunit
- common  $\alpha$  subunit (92aa)
- unique  $\beta$  subunit (118-121aa)

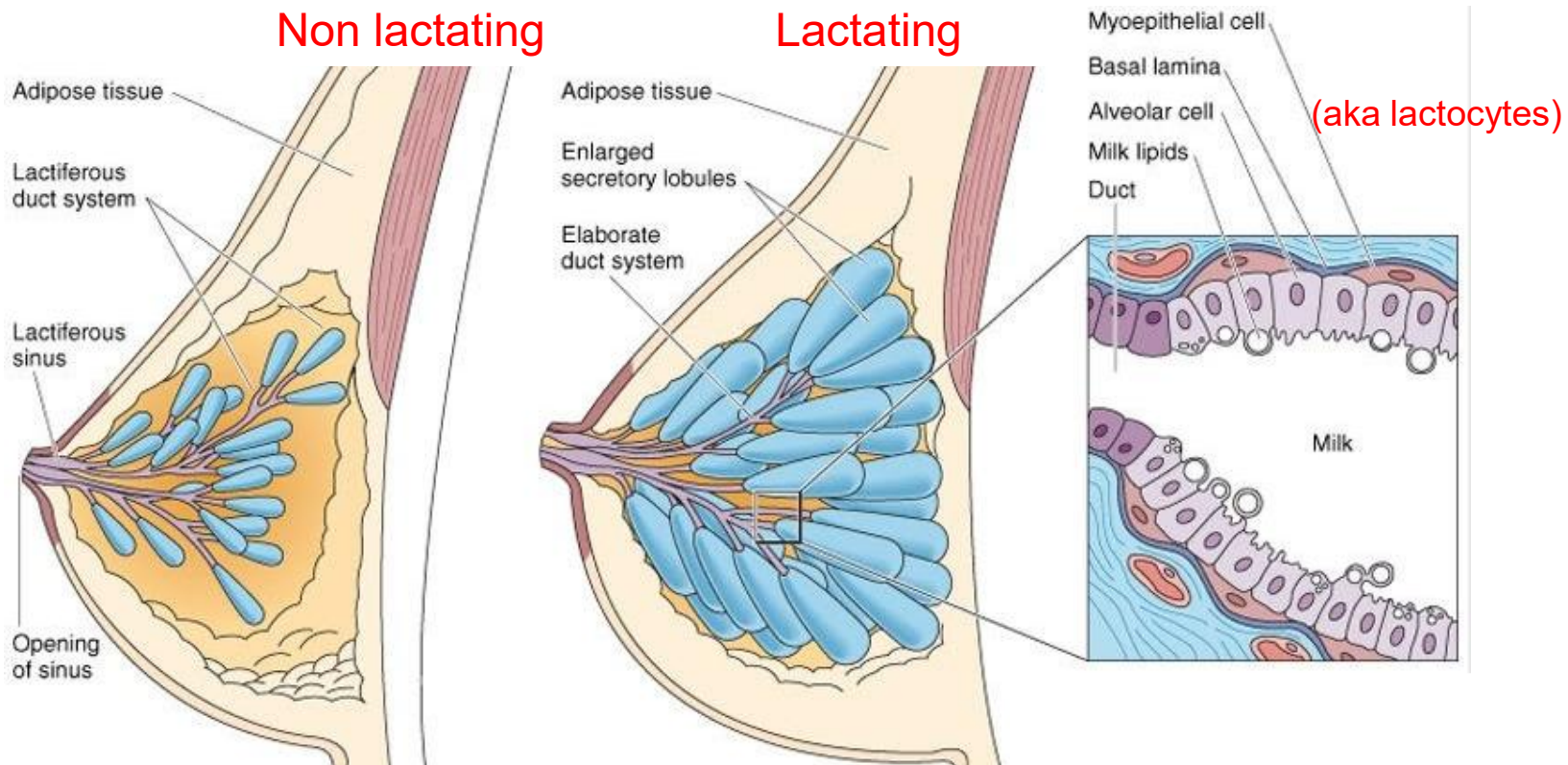
# Prolactin

- Somatomammotrophin
  - only major anterior pituitary hormone that is **not** a trophic hormone
- Regulation
  - estrogen/progesterone  $\uparrow$  ( $\uparrow$  prolactin levels during pregnancy)
  -  • prolactin release is **inhibited by dopamine** ('suckling reflex')
  - 1 (as yet unidentified) prolactin releasing hormone (PRF)
- Major physiological role is in **lactation**
  - immune tolerance ?
  - behavioural role
    - nurturing instinct
    - broodiness

# Prolactin Plays an Essential Role in Lactation

Prolactin is one of several hormones that stimulate the development of the breast as a glandular tissue during pregnancy

It is also necessary for breast milk production



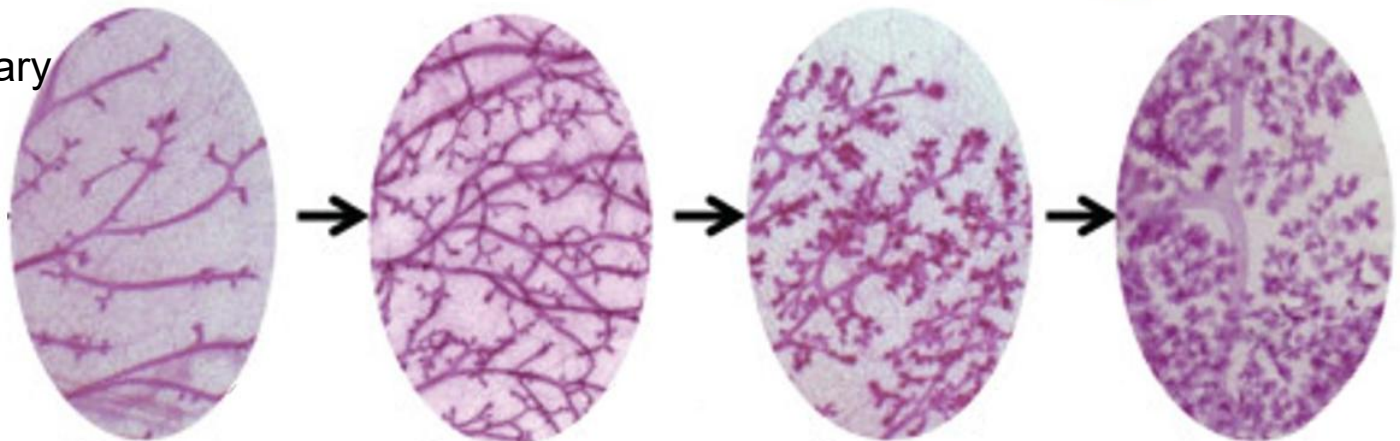
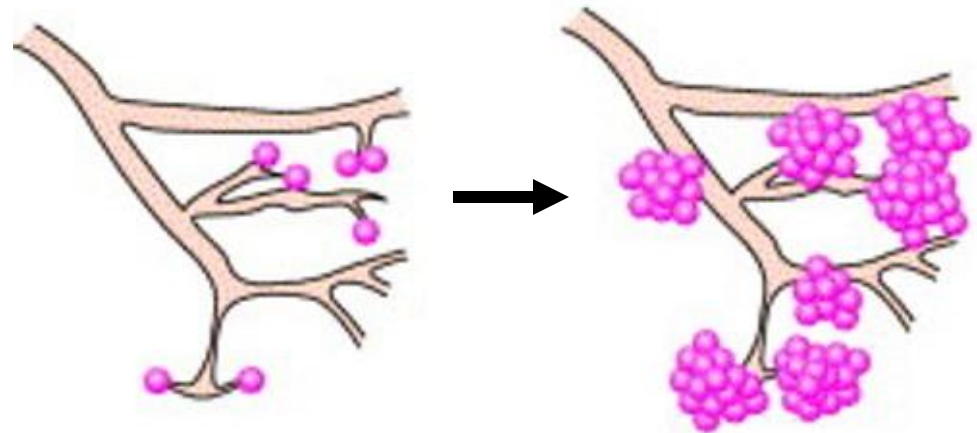
# Development of the Lactating Breast

Mammogenesis – development of the **functional** breast

Begins in puberty due to increased estrogen and progesterone levels and is completed in third trimester of pregnancy in response to sustained high levels of these hormones as well as prolactin

Prolactin specifically promotes lobuloalveolar proliferation and development during latter half of pregnancy

Prolactin is also necessary for milk production (**lactogenesis**) pre and postpartum



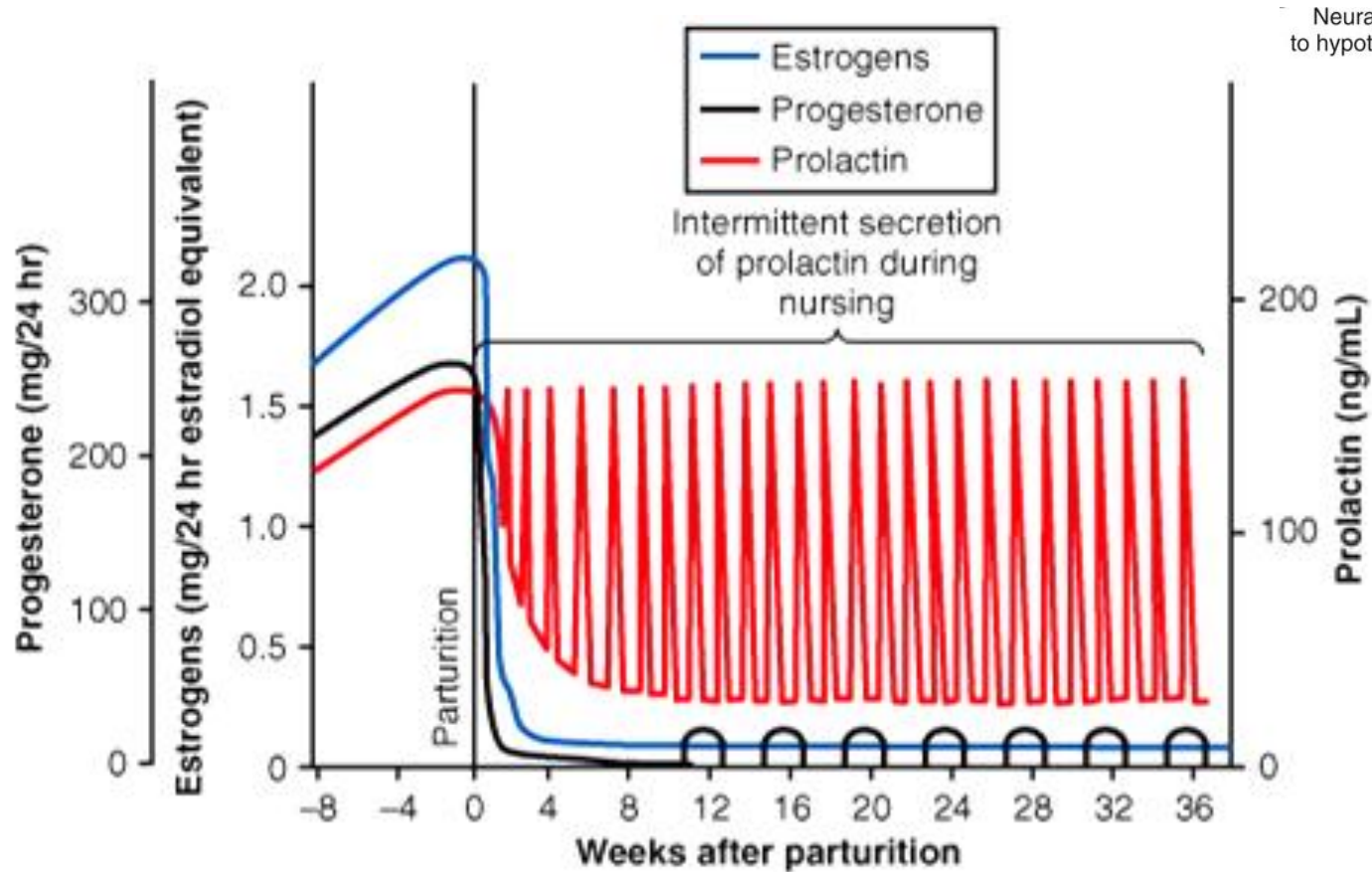
Non pregnant

Early pregnancy

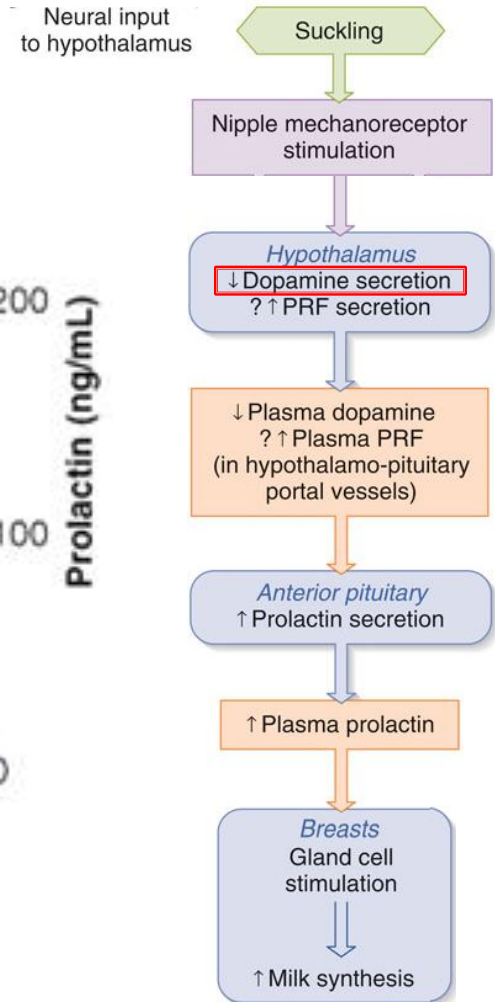
Late Pregnancy

Post partum

# Postpartum Lactogenesis

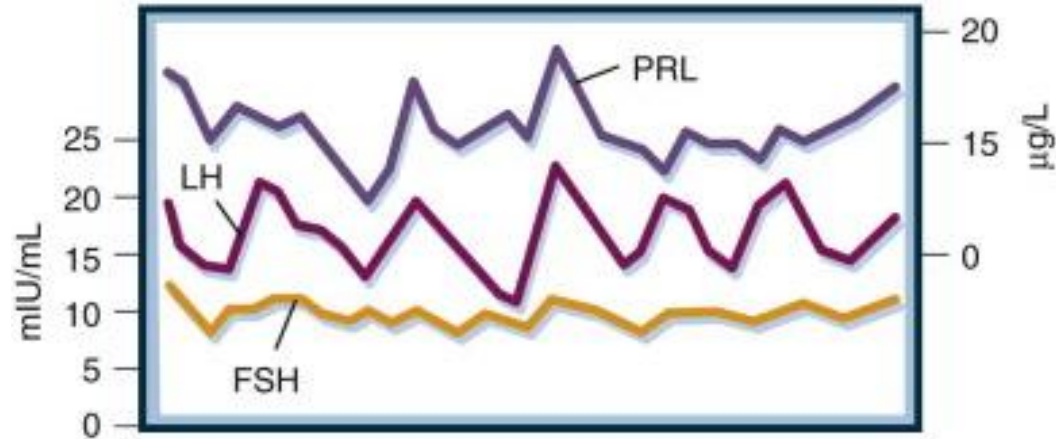


maintains ability to lactate (produce milk)

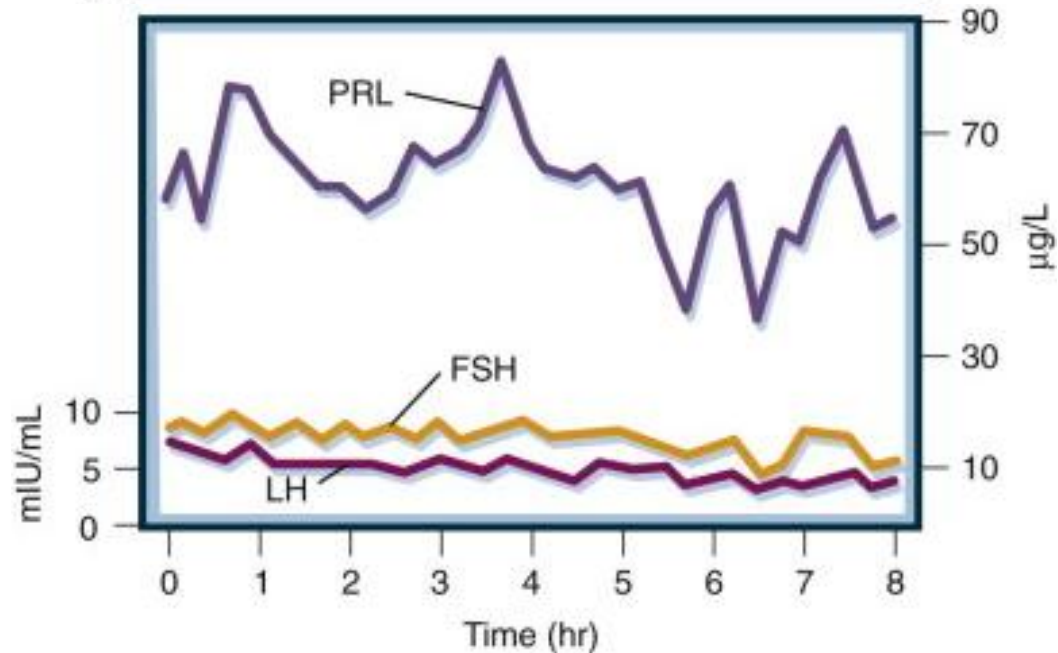


# Lactational Amenorrhea

normal hormone  
profile



hyperprolactinemia



# Hyperprolactinemia

- 40% of all cases due to benign (functioning) pituitary adenomas (prolactinomas)
  - other pituitary tumours or pathologies that lower dopamine levels in hypophyseal portal system ie. disinhibition
- Hyperprolactinemia stimulates lactation (milk production) and interferes with gonadal function (hypogonadism)
- Galactorrhea
  - occurs in about 80% of women, uncommon in men
- Amenorrhea
  - about 20% of cases of **secondary** amenorrhea due to hyperprolactinemia
  - inhibiting hypothalamic GnRH production with knock on effects on pituitary LH and FSH production and gonadal estrogen production
  - infertility and loss of libido is most common presentation of hyperprolactinemia in men

# Next week

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Tuesday 22nd Sept @ 11:00  
Endocrine Histology CAL D114/115

Wednesday 24th Sept @ 17:00 in B004 Health Sciences  
Building – GH/IGF-I Axis